

Justin Arackel

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Work Experience

Functional Hardware Verification Engineer

IBM

Austin, TX

June 2022 – Feb 2025

Verification/QA Work:

- Verified behavior for VHDL Logic for the following Circuits (CTLE, VGA, Servo Op Interface, PLL, ECC, FFE) within the IoPHY for P, Z and Sentient
- Participated in design reviews and contributed to RTL-level debugging and timing analysis.

Dev Ops Work:

- Maintained internal tooling infrastructure, including tooling updates and automation scripts (e.g., BASH, TCL, Perl, Python, CMake).

Additional Skills:

- Completed internal training and self-guided coursework on NLP and large language models and gained hands-on experience with foundational transformer architectures (Watson).
- Agile workflow - Professional expectations and presentation guidelines for meetings
- Created comprehensive documentation supplemented with custom visuals to enhance clarity and user understanding.

Software & Embedded Systems Engineering Intern

BEI Precision Systems & Space

Maumelle, AR

May 2021 – Aug 2021

- Test, Calibrate, Debug & Validate encoder in Lab
- Fill gaps in Design Validation and Document at the module level for the FPGA
- Develop initial simulation environment to investigate improvements to the current Processor and changes to Actel's SRAM IP

Education

Texas A&M University

BS in Electrical Engineering

Sept 2018 - May 2022

- GPA: 3.7/4.0
- **Coursework:** Ultrasound Imaging, Digital Image Processing, DSP Based Motor Controls, Microprocessor System Design, Digital Communications, FPGA Information Processing Systems, Digital Signal Processing, Electronics, Computer Architecture & Design, Fundamentals of Electromagnetism
- **Projects:** PVC Pipe Speaker: Developed Power Supply, Active Crossover and Power amplifier electronics and PCB associated with driving a Speaker/Tweeter System.

Skills:

Programming Languages: C++, C, Python, Forth, MATLAB, Verilog, VHDL, Linux

Tools: RTL Tools(Xilinx, ModelSim, Cadence), PCB Layout & Circuit Sim(Altium, MultiSim)